

Richard Álvarez

Chicago, Illinois ◦ rawalvarez731@gmail.com
richardalvarez.info ◦ GitHub ◦ LinkedIn

Research Interests

Human-centered artificial intelligence; human-computer interaction; reliable tool-using AI systems; computational film and media; multimodal creative tools; and AI-assisted media production.

Education

Kenyon College

Bachelor of Arts in Film

Aug 2020–May 2024

Gambier, Ohio

- Minor in History; concentration in the Integrated Program in Humane Studies.

Publications and Posters

- **Álvarez, Richard Anthony.** *A Retrieval-Augmented Film Recommendation System*. Poster, IPHS 484: Senior Seminar, Paper 34, Digital Kenyon, 2024. digital.kenyon.edu/dh_iphs_ss/34
- **Álvarez, Richard Anthony.** *Captivating Video Editing Secrets: Unsupervised Deep Learning and PySceneDetect Analysis of Creative Strategies*. IPHS 484: Senior Seminar, Paper 22, Digital Kenyon, 2023. digital.kenyon.edu/dh_iphs_ss/22

Professional Experience

AI Solutions Engineer

Layer 7 Systems, Independent Contractor

Sep 2025–Present

Remote

- Built and operated a stateful TypeScript, Bun, and Express MCP service connecting Retell voice agents with Autotask and Microsoft Graph/Teams.
- Implemented identity and company resolution, contact-safe ticket access, authorization gates, ticket actions, technician routing, post-call processing, and human handoff.
- Developed an authenticated React operations dashboard, maintained a 756-test regression suite, and deployed Docker services through GitHub Actions.
- Analyzed 60 workflow recordings from 14 users and identified 176 moments suitable for AI assistance.
- Built a Next.js and Neon/PostgreSQL learning platform for 34 trainees across four courses and 16 modules.

Freelance Software and Web Developer

Richard Álvarez, Sole Proprietorship

Oct 2024–Present

Chicago, Illinois

- Design and deploy responsive applications using Next.js, React, Node.js, Tailwind CSS, Stripe, PostgreSQL, authentication, and object storage.
- Scope, document, deploy, and maintain client applications, hosting, and content workflows.

IT Assistant

Kenyon College Library and Information Services

Sep 2023–Feb 2024

Gambier, Ohio

- Supported workstation imaging, hardware deployment, classroom technology, and recurring maintenance scheduling.

Selected Software and AI Projects

Vigil

2025–2026

Half-Time Draw Predictor

2026

Chroma

2026

GREasyVocab

2024

Bitset Game of Life	2024–2026
Kubrick-Style Script Generation	2024
Rotten Tomatoes Sentiment Analysis	2024
DV Tools	2025
Business and AI Training Platform	2026
Workflow Capture and Analysis Pipeline	2026
Configurable Voice-Reception Template	2025
Message-Box Auto-Attendant	2025
VibeVoice Evaluation Prototype	2025
Selected Professional Projects	
Joaquin Morales Cinematography Portfolio	2024–2026
Machine Television	2024–2025
BDPA HSCC Website Prototype	2025
Personal Portfolio	2024–Present
Professional and Community Service	
High School Computer Competition Judge	2024–2025
<i>BDPA</i>	<i>Chicago, Illinois</i>
Reviewed and scored student programming submissions for the 2024 and 2025 national competitions and completed formal judge training.	
Professional Certification	
CompTIA IT Fundamentals (ITF+)	2024
Technical Skills	
Languages and application development: TypeScript, Python, JavaScript, SQL, C++ , HTML/CSS, Shell, Bun, Node.js, Express, Hono, FastAPI, React, Next.js, Electron, and Zod. Data and applied AI: PostgreSQL, Neon, SQLite FTS5, MCP, REST APIs, tool calling, structured outputs, retrieval-augmented generation, context engineering, OpenAI, Anthropic, Gemini, and Retell. Integration and operations: Autotask, Microsoft Graph/Teams, Docker, GitHub Actions, Linux/VPS operations, CI/CD, automated testing, validation, structured logging, observability, and incident response. Computational media: PySceneDetect, FFmpeg, NLP and sentiment analysis, computational film analysis, and interactive media.	